

"Database Management System CS403"

Lecture # 1

Database:-

The place where data is stored or organized is called Database. To Manipulate (enter, delete, select, edit etc) the database, Languages are used such as Structured Query Language (SQL)

DBMS:-

On the other hand, Database Management system is a program or tool to manage the database and its users. DBMS & Database are jointly called Database systems.

Lacks of file systems:-

- 1) Program and Data interdependence
- 2) Non-sharing of data, separate files for each entity
- 3) Redundancy & wastage of storage
- 4) Inconsistency
- 5) Difficult Manipulation

Advantages of Databases:-

- 1) Data Sharing, use shared system for all entities.
- 2) Program and Data Independence
- 3) Controlled Redundancy
- 4) Better data Integrity, validation of data
- 5) Easy Manipulation

CS403 Lecture # 2.

Data:-

Data is collection of raw facts & figures. ~~is called~~. It is collected from any specific environment for a specific Purpose.

Information:-

The processed form of data or meaningful data is called information.

Schema:-

It is a structure to express the format and other information about data & Database.

Database Application:-

It is a program or set of programs that are used for performing operations on data stored in database.

DBMs:-

It is a Database Application/software.

Two basic operations performed by DBMS are

- 1) Management of data in database
- 2) Management of users associated with database.

Advantages of DBMS:-

- 1) Data Consistency
- 2) Data Security
- 3) Economy of scale
- 4) Concurrency control
- 5) Backup & Recovery
- 6) Faster development of new Applications according to growing future needs.

* High cost and Conversion cost are two costs in Adopting a DBMS for any company.

Data as a Resource:-

Data is considered as a Resource for an organization to make proper decisions at a proper time for success of its business. It is only data which provides correct information to make proper decisions.

Levels of Data:-

Real world data: Data about any real object

Meta Data: Data about data, properties of object

Existence of data: Actual data

Database Users:-

There are Two types of DB Users

Application programmers:-

Application programmers are the skilled people who create DB Programs according to needs of users of DB.

End users:-

People who use database for their Personnel working are End users. End users are further classified into two types.

1) Naive users: who simply use database

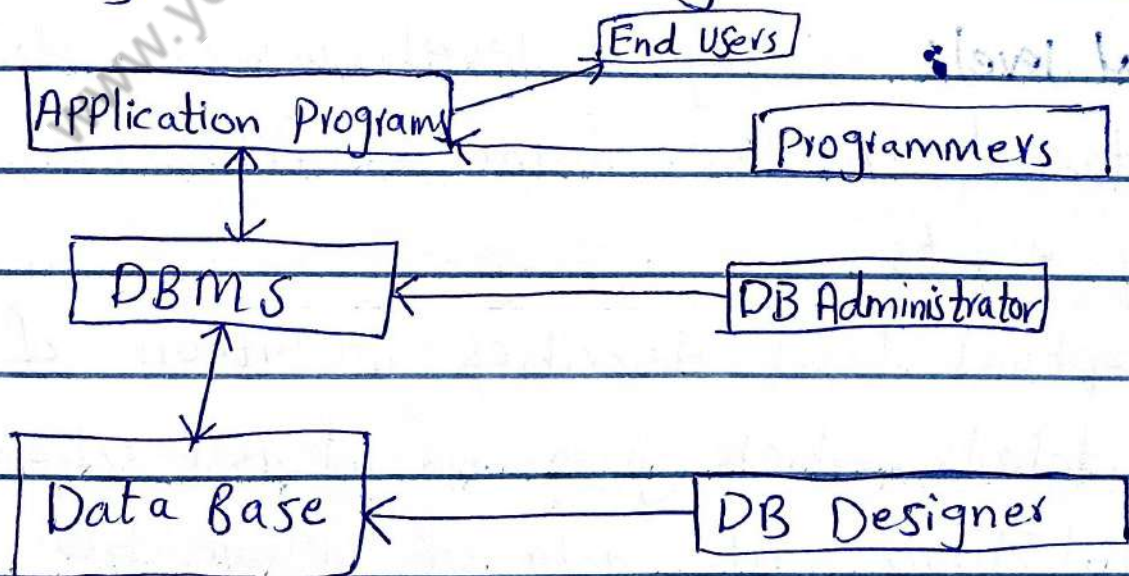
2) Sophisticated users: They use database as well as given extra rights to access and manage data.

DB Administrator:-

This class of dB users are most technical class. They know how to design & manage the database as well as to manage data in database.

Duties of DBA:-

- 1) Routine maintenance
- 2) Granting data access
- 3) Schema Definition
- 4) Making proper data Backups
- 5) Recovering dB in case of any crash



Lecture # 3

Database Architecture:-

The structure of DB is called dB Architecture. DB Architecture was proposed by ANSI SPARK in 1975. It is three level architecture which hides details of internal systems.

External level or View level is only related to user. User access data and see it in External View level.

Conceptual level or Logical level where data is stored in the form of tables.

Internal level or Physical level where data is stored in the form of files as binary digits.

★ Conceptual level describes intention of dB.

★ The details which gives us internal schema are Intentions while data of organisation is its Extension. Extension is performed on

basis of complete intension.

Name : Ali
Class : BSC
CGPA : 3.95
Roll No : 1

→ External or
View level/schema

Name	class	CGPA	Roll No
Ali	BSCS	3.95	1
Asim	BSIT	3.34	2
Asif	BS.M	3.88	3
//	//	//	//
//	//	//	//

→ Conceptual
or
Logical level
schema

C1	C2
C3	C4
C5	C6
C7	C8

101000

→ Internal/Physical
Level or schema

Lecture # 4

Data Independence:-

Data Independence is a major objective of 3-level architecture. It means changes made to data does not effects Program's working and vice versa. It is classified into two types.

Logical data Independence:

It means changes made in conceptual level do not effect the external/view level.

Physical data Independence:-

It means changes made in Internal/Physical level do not effect the conceptual level.

Functions of DBMS:-

- 1) Data Processing
- 2) Integrity
- 3) Concurrency
- 4) Transaction support
- 5) Backup & Recovery
- 6) Data Communication

7) Authorization services

8) User Accessible catalog

* User management & Data Management are two Basic functions of DBMS.

DBMS Environments:-

There are two types of environment

Single user :-

DBMS might have several number of users but only one user can access to DB system at a time. This type of DBMS is also called Desktop Database system.

Multuser :-

This type of DBMS allows multiple users to access DB system at the same time.

This environment has 3 types:

Teleprocessing:-

In this type, all users are

requesting to central computer where DB is residing. The results are transported back to users. This type is obsolete or outdated now.

File servers:

In this type, the server acts as a storage device. Each user/client has its own copy of DBMS software. To access even single record, the server sent entire db file to client. After editing, the client sends the entire file back. It causes network overhead.

Client Server :-

A server runs the DBMS, while clients run only application software. The clients send specific query/request. The server processes data internally and sends back only the result. It is highly efficient and mostly used technique.

Lecture # 5

DB Development process:-

1) Preliminary Study:-

It means initial or proper study of system which is to be developed.

2) Requirement Analysis:

In this phases, we conducted a detailed study to find out requirement of system

3) Database Design:-

This is a phase where conceptual or logical design of database is created.

4) Physical Design:-

In this phase, the conceptual design is converted into physical design by implementing database into a specific DBMS.

5) Implementation:-

In this phases, we write different applications programs needed to carry out

activities according to user's requirements.

6) Maintenance:-

It means to fine tune the system and check that designed applications systems are doing their work properly.

Data flow Diagrams :-

It is a most common tool used for designing database. DFDs Graphically expresses the flow of data in DB.

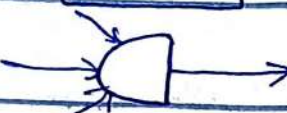
Advantages:-

- 1) DFDs are simple and Hide complexities.
- 2) DFDs are descriptive.
- 3) DFDs show flow of data b/w processes.

Limitations:-

- 1) They do not Provide us a way of expressing decision points.
- 2) They focused on flow of information only.

Symbols in DFDs:-

- 1) Data flow 
- 2) Data Store 
- 3) Process 
- 4) External Entities 
- 5) Collector 
- 6) Separator 
- 7) Ring-Sum operator 
- 8) AND Operator 

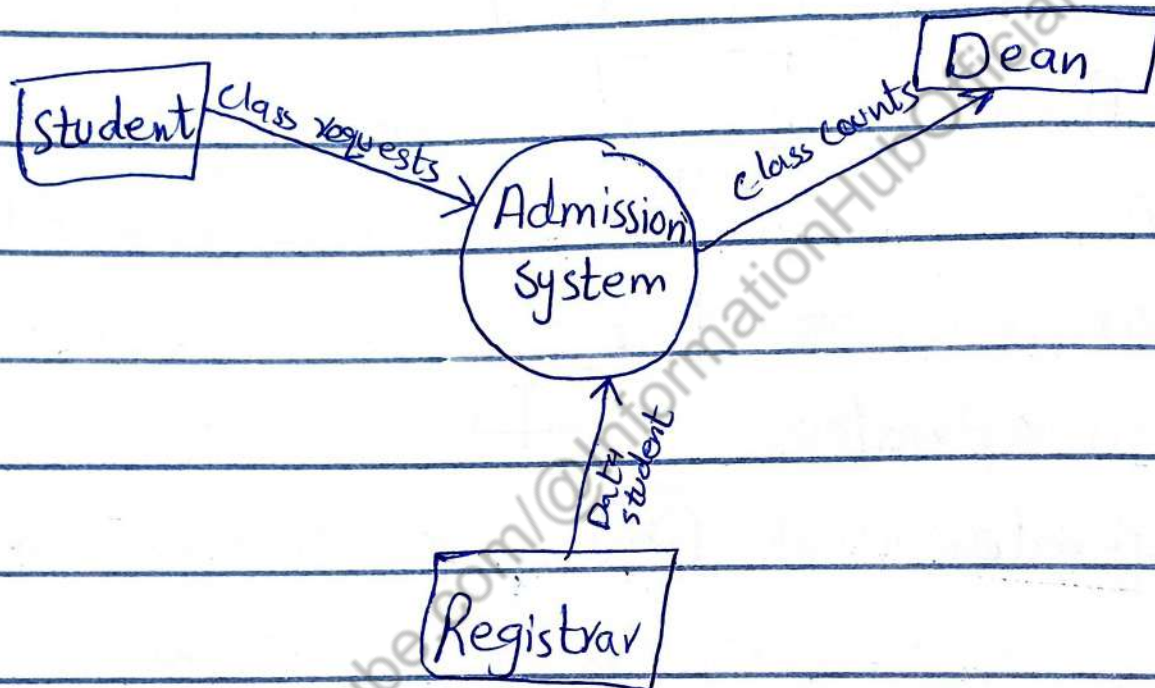
Types:-

- 1) Context Diagram
- 2) Level 0 Diagram
- 3) Level 1 Diagram or Detailed Diagram

Context Diagram:-

It provides least information about working of a system. They always consist of a single process and shows its interactions with external entities only.

No data stores were created and no communication b/w external entities is shown.



Level 0 Diagram:-

This diagram is bit much detailed than context level Diagram. It also uses only three DFD symbols like CDFDs, but it interacts with large number of external entities and processes.

Lecture # 6

Level 1 Diagram:-

It is more detailed representation of a system than Level 0 Diagram. It breaks its single process into multiple sub-processes. The sub-processes are given number like 1.0, 1.1, 1.2 etc. for clarity purpose, there should be 7 to 9 processes in 1 detailed diagram.

Data Dictionaries:-

It is a database that contains data about all databases in database system. There are two types of data dictionaries.

Integrated:-

It is a dictionary which is placed embedded into database system, and is created by DBMS for its usage in different stages.

free standing:-

The dictionary which is created

by any case tool and then attached to Database system.

Gross Reference Matrix:-

It is two dimensional array used for finding entities & their associations in database. Its rows or x -axis specifies the attributes and its columns or y -axis contains different objects, reports or transitions. The (✓) Tick mark in a cell indicates the presence of a specific attribute in given object/report.

	Women	Fish
Swim		✓
walk	✓	
Eat	✓	✓

Database Modeling:-

The Process of creating logical structure of database is

called database modeling.

Data Model:-

It is set of constructs used for creating a database and producing design for database. Few components of data model are

- 1) structure / ^{construct} : for storing data
- 2) Manipulation language : for manipulating data
- 3) Integrity Constraints: Ensures correctness of data.

Types:-

* Semantic data models provides flexibility for expressing complex relationships. Its types are

- 1) ER-data Model
- 2) Object-Oriented data Model

* Record based data Models provides logical structure of database. Its types are

1) Hierarchical data Model

2) Network data Model

3) Relational Data Model

* No DBMS is based on ER data model because it is purely used for designing.

Types of Database Design:-

- 1) Conceptual Database Design
- 2) Logical Database Design
- 3) Physical Database Design

Lecture # 7

ER-Data Model:-

It is a semantic data model used for graphical representation of conceptual database design. It has three constructs:

- 1) Entity
- 2) Relationship
- 3) Attribute

Entity:-

Entity is any real object that has some attributes/Properties. Following three terms are related to an Entity.

Entity type:-

~~It means the specific object about which data is belonging.~~ It is defined as a Name/label assigned to ~~a~~ objects that have similar attributes.

* The process of identifying entity types, their properties and relationships b/w them is called Abstraction.

Entity Instance:-

It means an object ~~is~~ belongs to which entity type. It is an instance of an entity type. For Example, Student is Entity type and a Particular Student "Ali" it is Entity Instance.

Entity Set:-

A Group of Entity instances of a particular entity types is called entity set. For Examples All students of a class forms an Entity set.

* The word Entity is used for all these three terms mentioned above.

* Entity types are classified into two types
1) **Strong Entity type:-** that can exist independently. They are represented by single rectangular box

Student

Weak Entity type:- that can't exist independently. They need to link to instances of other entity types. They are represented by Double line rectangle

Dependents

Attribute:-

The properties/Qualities of a specific entity instance are called Attributes. Entity instances of same entity type have same Attributes but attribute's values can be different. Student in an Entity & its name, age, Gender are its Attributes.

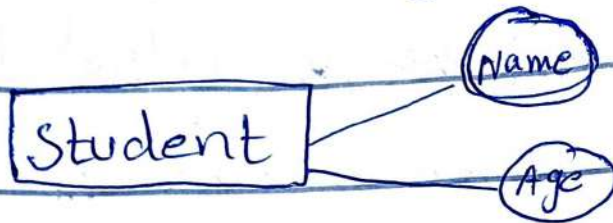
Domain of Attributes:-

The possible values that an attribute can have. Domain is form of check/law on attribute so that it can't have value outside this set. It helps in maintaining integrity of database

★

Attribute

is represented by an oval linked with an Entity.



Types of Attributes:-

Simple Attributes: Attributes are of single E-type

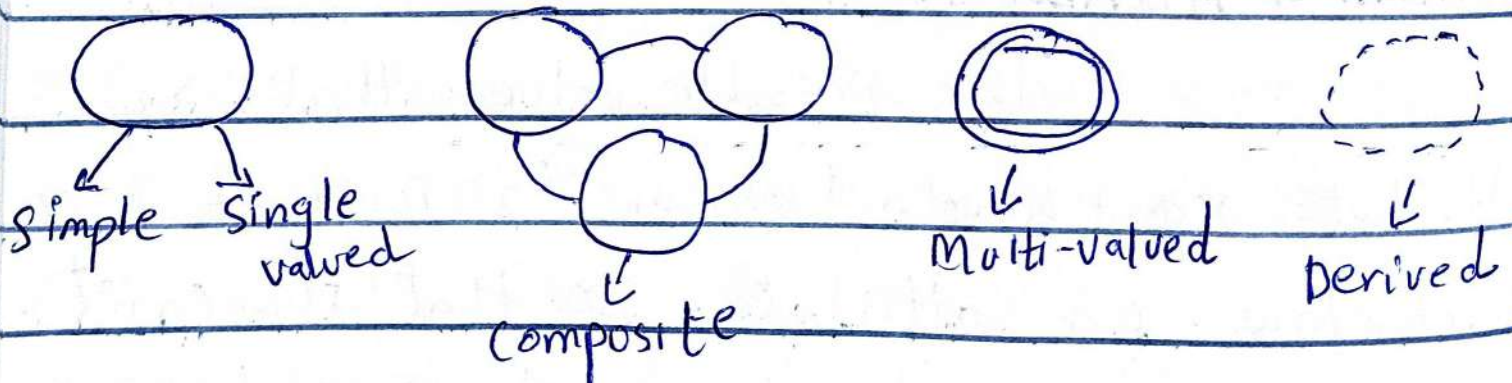
Composite Attributes: Attributes are from multiple E-type

Single-valued: Have only single value

Multi-valued: have multiple values (Hobbies)

Derived: whose values derived from other Attributes.

For Example: Age can be derived from DOB.



Lecture # 8

The keys:-

A key is a set of Attributes that can be used to identify or access an Entity instance of any entity type.

Types:-

1) Super key:-

It is a set of 1 or more attributes that can uniquely identify an entity instance in entity set. Any combination of attributes with the super key is also a super key.

2) Candidate key:-

A super key for which no subset is a super key is called a candidate key. or a minimal super key is candidate key.

* Every candidate key is super key but not every super key is candidate key.

Primary key:-

Its definition is same as candidate key but a table can have only one primary key. It can't contain null values and values must be unique.

★ Every Primary key is a candidate key.

Alternate key:-

Candidate keys which are not chosen to be primary keys are called Alternate keys.

Foreign key:-

Attributes

A column or set of columns that refers to PK in another table is called foreign key. It is a link b/w two tables which enforces referential integrity.

Secondary key:-

An attribute or set of

Attribute that can access values of an entity type without the use of PK. It may not return unique values.

- 1) ID
- 2) Name
- 3) SSN
- 4) Email
- 5) ~~Name~~ Father-ID
- 6) Address

Candidates are ID, SSN, Email

Primary key is ID

Alternates are SSN, Email

Foreign key is Father-ID

Secondary key is Name + Address

Super key $\{ID + Name\}$, $\{SSN + Email\}$,
 $\{ID\}$, $\{Email + Name\}$

Lecture # 9

Relationship:-

A relationship is an association or linkage between two entities of interest.

Each relationship has name, ^{optionality} and degree.

Relationship is represented by a diamond symbol b/w entities. Diamond is doubled if one entity is dependent on other.

Types:-

- 1) Unary: relationship with itself
- 2) Binary: relationship with two
- 3) Ternary: relationship with three
- 4) N-ary: relationship with N entities

Relationship Cardinalities:-

Cardinality means the number of times the one entity is linked to other entity. It has four types:

One to One:

Example: Student have roll NO

One to Many:-

Example: Hand has fingers, student has books

Many to One:-

Example: subjects in class,

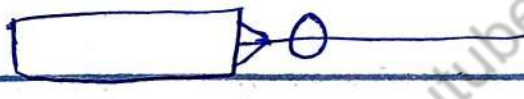
Many to Many:-

Books have Readers



 Optional One

 Mandatory One

 Optional Many

 Mandatory Many

* A relationship type is an abstraction of a relationships & entities enrolled in relationship are called its participants.

Lecture # 10

Other Notations:

- 1) Arrow Head Notation: 
- 2) Alphabetical Notation: 
- 3) Dot Notation: 

Role:-

The way an entity is enrolled in a relationship is called its role.

Recursive Relationship:-

It is a relationship of an entity with itself. The link initiates from one entity and terminates on same entity. This is basically a unary relationship.

Dependencies:-

It is a constraint which means how one entity is dependent on other. It has 3 types.

Existence Dependency:-

When one entity couldn't exist

without the presence of other entity, it is called existence dependency. Example: fish entity can't exist without water entity.

Identifier dependency:-

It means the dependent entity has not its identifier and uses any external identifier to pick its data.

Referential dependency:-

When primary key of one entity is referred to other entity in order to approach its attributes, it is called referential dependency.

Lecture # 11

Super Type:-

It is a parent class or entity having specific characteristics.

Sub Type:-

It is a derived/child entity having all characteristics of its Parent entity and also some unique properties. Subtypes hold all properties of their corresponding super types.

Inheritance

The Transfer of characteristics from super type to subtype is called inheritance. With its help, we don't need to write attributes again & again.

✶ After inheritance, there are a number of constraints for specifying further restrictions on relationships:

Total Completeness Constraint:-

When all attributes of

super type should present in subtype, it is called total completeness constraint.

Partial Completeness Constraint:-

When some attributes of parent table should present in subtype, it is called partial completeness constraint.

Disjoint Constraint:-

When there are more than 1 child or subtypes of super type, then only one child can have characteristics of parent/supertype at one time.

Overlap Constraint:-

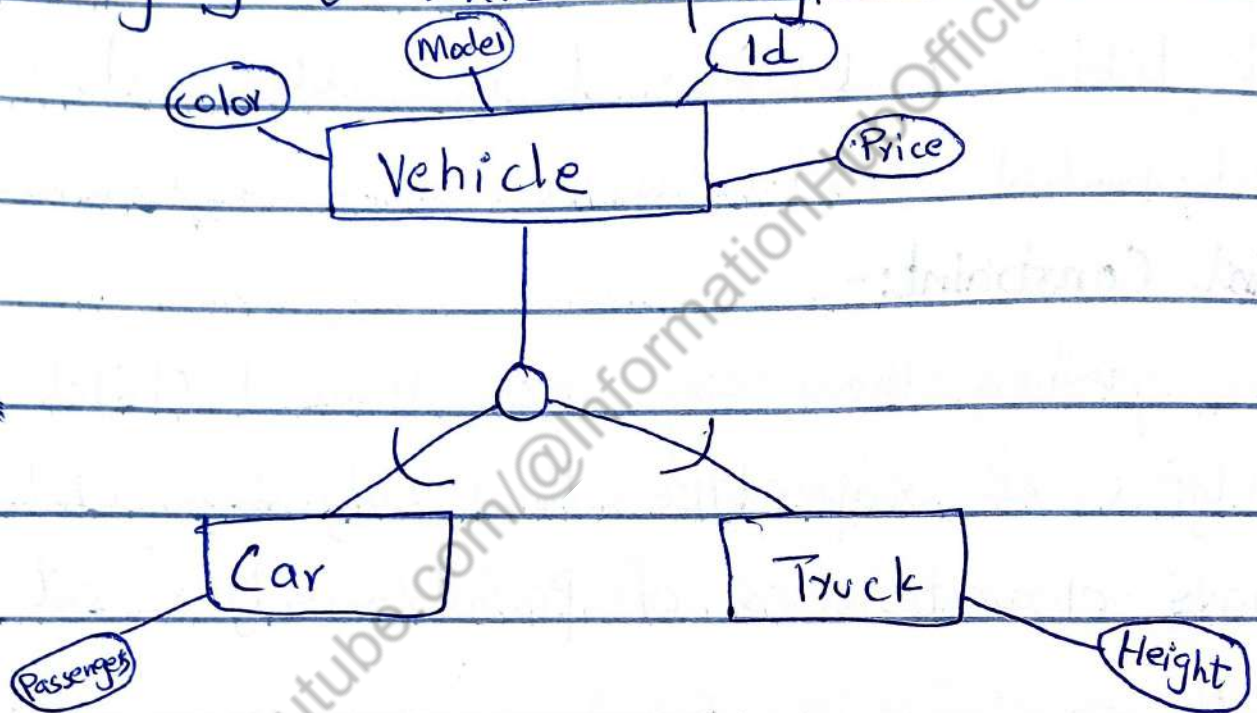
This is contrast to disjoint rule, it allows multiple subtypes to have characteristics of supertype at the same time.

Types:-

- | | |
|-------------------------|------------------------|
| 1) Complete Disjoint | 3) Partial disjoint |
| 2) Complete Overlapping | 4) Partial overlapping |

Subtype Discriminator:-

It is a technique which is used to know that which subtype is belonging to which supertype.



Lecture # 12

This lecture is only practical. Drawing conceptual level, level 0 and level 1 diagrams. we cover these topics in practical short lectures so we are skipping it for now.

Lecture # 13

Cardinality & Relationship b/w Entities:-

Program & Courses:-

The relationship b/w program & courses is One to many (1- $*$), because one program can have multiple courses. The relationship b/w courses & program is many to Zero ($*$ -0), because ~~a~~ many courses can have even no program as well.

Students & Programs:-

The relationship b/w students & program is One to One (1-1), because one student can have only one program at a time. The relationship b/w program & student is many to zero because many programs have no students ~~yet~~ as it is not offered yet.

Semester & course:-

The relationship b/w semester & courses is many to many because ~~many~~

multiple semesters are running in a university that have many courses.

Course & Teacher:-

The relationship b/w course & Teacher is One to One because One course has only one teacher. The relationship b/w Teacher & course is one to many because One Teacher can teach multiple courses/ subject at a time. Like me 😊

Student & Course:-

The relationship b/w student & course is many to many because many students have multiple courses at a time.

Semester & Student:-

The relationship b/w semester & students is also many to many because many ~~semester~~^{or} multiple semester in university can have multiple students at a time.

Lecture # 14

Question:-

Difference b/w Conceptual Database design & Logical database design?

Conceptual DB Design

Developed in a semantic data model (ER Model)

2) It results from analysis Phase.

3) It represented Graphically.

4) More Expressive

5) First transformed & then Implemented.

Logical DB Design

Developed in Relational data Model (RDBMS)

It obtained by Translating Conceptual DB design into another model

It represents data in descriptive form.

Less Expressive

Going to be implemented.

Relational Data Model:- & RDBMS:-

An IBM Scientist E.F Codd

Proposed RDM in 1970. Before that, all database systems are based on two older

data models which are hierarchical data model & network data model. Relational Database Management system were developed by IBM & UC-Berkeley by the mid 70-s, which provides foundations for other leading DBMS products like Oracle, sybase, informix, Microsoft Access and SQL Server.

RDM Advantages:-

- 1) RDM provides simplicity as there is only one structure that is a table or Relation.
- 2) It gives very strong Mathematical foundation.

Table/Relation:-

A two dimensional structure is called Table. The two dimensions are rows & columns. Rows of a Table are called Tuples and columns are called

Attributes or Properties.

* The intersection of a row & column is called Intersection.

RDBMS Advantages: Properties:-

- 1) Each cell of a table contains single / Atomic value.
- 2) Every column name is unique.
- 3) The values of Attributes come from the same domain.
- 4) The order of column is immaterial i.e., doesn't matter.
- 5) The order of rows is also immaterial.
- 6) Each row is distinct, no two rows can be same.

Components :-

- 1) Structure
- 2) Manipulation Language
- 3) Integrity Constraints

Relation:-

IF $A = \{x, y\}$, $B = \{2, 4, 6\}$, then

$$A \times B = \{(x, 2), (x, 4), (x, 6), (y, 2), (y, 4), (y, 6)\}$$

A Relation is some subset of this

Cartesian product. Like $R_1 = \{(x, 2), (x, 4), (x, 6)\}$

Example:-

STD Relation = $\{(S0, Ali, 12/12/76), (S1, Asim, 2/12/77)\}$

The above relation can be represented in a two-dimensional structure or Table.

ID	Name	DoB
S0	Ali	12/12/76
S1	Asim	2/12/77

Lecture # 15

Math Relations Vs RDB:-

The order of columns in mathematical Relations matters but in RDB, it doesn't matter. For Example, in Mathematical Cartesian product

$$A \times B \neq B \times A$$

but for RDB $A \times B = B \times A$

* Rest of Properties b/w them are same.

Degree of a Relation:-

The number of columns in a relation is called Degree of a Relation

Cardinality of Relation:-

The number of rows in a relation is called Cardinality of a relation.

Foreign key:-

An Attribute of Table B that is Primary key in another Table A is called Foreign key.

EMP (empId, Name, qual, deptId) B Table

DEPT (deptId, Name, numEmp) A Table

Properties:-

- 1) The number of foreign keys in a table depend upon the relationship of this table with other tables.
- 2) The foreign key and its related primary key can have different names but must have same domain.
- 3) The foreign key is represented by dotted lines under attribute while primary key is represented by solid line.

Entity Integrity Constraints:-

This rule focuses on primary key of a Table. It says the primary key column cannot contain Null values.

Referential Integrity Constraints:-

This rule is applied to foreign keys. It says foreign key value must match with its related PK value or F.K value.

must be completely Null.

- * PK is a minimal identifier that is used to identify tuples uniquely.

- * FK Provides correctness, validity or integrity of database

Null Constraint:-

with its help, we can't give null values into cells of specified attributes.

Default value:-

It means the value which is automatically given in cell of specified attribute when user didn't give any value.

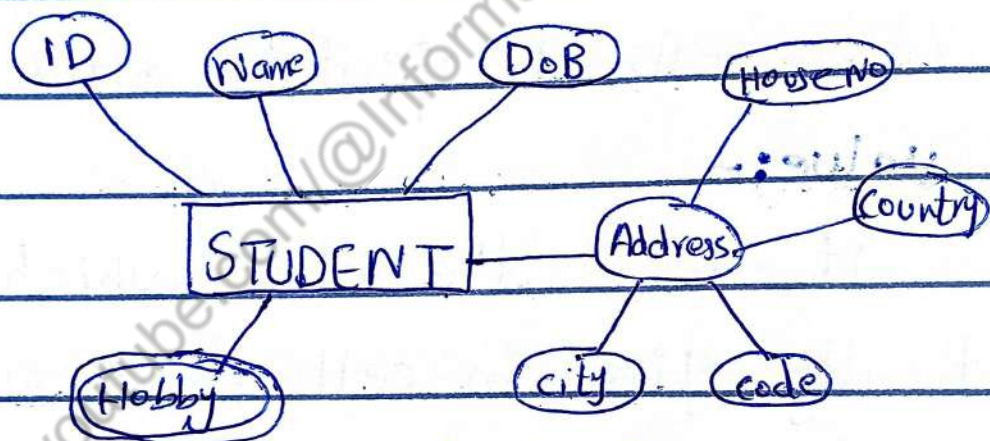
Domain Constraint:-

It means that every attribute has got a domain. Domain means the possible set of values that an attribute can have.

- * The Process of transforming Conceptual DB into Logical DB involves transforming of ER model

into Relational Model. There are two ways of transforming.

- 1) First is manually that is we analyze and evaluate and then transform.
- 2) Second is CASE Tools



Conceptual DM

STUDENT (ID, Name, DoB)

ADDRESS (House No, Country, code, city)

HOBBY (ID, hobby)

Logical DM

Lecture # 16

Relation is a structure created for an entity

Relationship is an association b/w two or more Relations.

Binary Relationship:-

when relationship is b/w two relations

It has further 3 types:

One to One:-

When only One Attribute of first relation is connected to one attribute of second.

For Example, One student can have only one scholarship at a time, so there is One to One relationship b/w these two Relations.

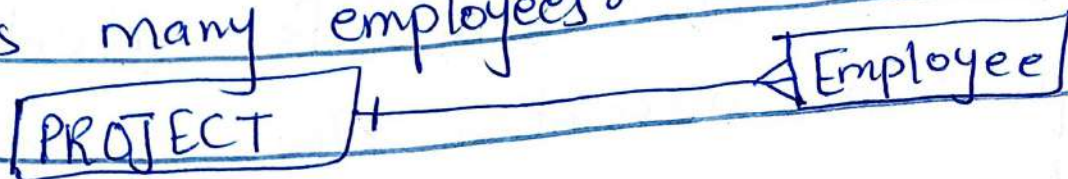
STDC (id, name)

STAPPLE (STid, Amount, id)

One to many:-

When One attribute of first relation is related to many attributes/instances of second relation. For Example, One Project

has many employees.



3 Many to Many:-

When many attributes of first relation are related to many attributes of 2nd relation. In many to many relationship, a third table is created which is also called Associative entity type which contains PK of first and 2nd relations. STUDENT & BOOK has Many to Many Relationship.

Unary Relationship:-

Relationship with itself is called unary relationship or Recursive relationship. It can have One to One, One to many, and many to many Cardinalities.

Discriminator:-

To Link a super type with its

subtype, there is a requirement of descriptive attribute which is called discriminator. It is used to identify which subtype is linked with which supertype.

* Common Attributes are assigned to supertype & specialized attributes are assigned to subtypes.

Manipulation Languages:-

Languages used to perform many operations like insertion, deletion and updation on data in database are called Manipulation Languages. It has two types.

1) Procedural Language:-

Those languages in which what to do and how to do on database is required are Procedural Languages.

2) Non-Procedural Languages:-

Those Languages in which only

what to do is required are Non-procedural languages. Example is SQL.

★ Relational Algebra:-

Relational Algebra operates on one or more relations to make or define another Relation.

Properties,-

- 1) It allows expressions to be nested just as in arithmetic expression. This property is called closure.
- 2) There are five Basic operations in relational Algebra, which are
 - 1) Selection
 - 2) Projection
 - 3) Cartesian product
 - 4) Union
 - 5) Set Difference.
- 3) It has also Join, Intersection & Division Operations which are expressed in terms of 5 Basic Operations.

Lecture # 17

Relational Algebra:-

Relational Algebra is a Procedural Query Language. Its operations are divided into two categories:

1) Unary Operations:-

Operations which involves in only one relation are Unary ~~rel~~ operations. These are Select (σ) and Project (π).

2) Binary Operations:-

Operations which involves in two relations are Binary Operations. These operations are Union ($\cup$), Set difference ($-$), and Cartesian Product ($\times$).

Select operation σ :

This operation is used to select certain rows (based on condition) from the relation. It is represented by Greek letter Sigma (σ). Its syntax is

$\sigma_{\text{Condition}}(\text{Relation Name})$.

Project Operation: π :-

Project operation is used to extract values of specified columns and put them in a new relation. It is represented by Greek symbol π (π). Its syntax is π selected columns (Relation Name)

* Both select & Project operations can also be used in nested forms.

Union Operation $\cup$:-

It is used to merge two relations.

Union of two relations is only possible when they have same attributes and domain of attributes. Its syntax is

Table 1 Name $\cup$ Relation / Table 2 Name

* The Union Operation is commutative

$$A \cup B = B \cup A$$

Intersection Operation:- $\cap$

It selects only the common attributes from both relations and put them in a separate relation.

* Intersection is only possible when both ^{tables} have same attributes and domain. Its syntax is
 $\text{Relation 1 Name} \cap \text{Relation 2 Name}$

* Intersection is also commutative $A \cap B = B \cap A$

Set Difference:- $-$

It eliminates all attributes which are common in both relations and then put only those attributes in new relation which are remaining in first table. Its syntax is

$\text{Table 1 Name} - \text{Table 2 Name}$ (Not commutative)

Cartesian Product $\times$:-

It will merge each row of first table with each row of second table. Its syntax is

$\text{Table 1 Name} \times \text{Table 2 Name}$

The resultant table has $m \times n$ columns and $C \times D$ rows. This cross product is also commutative. $A \times B = B \times A$

Exercise "

Midterm syllabus Complete"

Lecture # 18

Joins:-

Join is a special form of cross product of two tables. It is a binary operation. It has 5 types.

1) Theta Join ($\bowtie \theta$):-

We select rows based on attributes from one table and then make cross product with each row of 2nd table. It is denoted as $R \bowtie \theta S$

2) Equi Join ($\bowtie =$):-

We join tables based on common attributes b/w them. We didn't take cross product after. $R \bowtie = S$

3) Natural Join:- $\bowtie$

This is most common and general form of joins. Its working is similar to Equi join but the common attributes appears only once. $R \bowtie S$

4) Semi Join $\bowtie$:-

We take Natural join first and then add attributes of first table only in new table.

$R \bowtie S$

5) Outer Join: $\Join$

Outer join returns matching rows

between two tables as well as the non

matching rows from one or both tables.

To Understand this, first we have to study its types.

Left Outer Join: $\Join$

Returns all rows from left table

and only matched rows from right table.

Right Outer Join $\Join$:-

Return all rows from right table and

only matching/common rows from left table.

Relational Calculus:-

It is non-procedural language

used to retrieve data from tables. It

has two forms:

- 1) Tuple Oriented Relational Calculus
- 2) Domain Oriented Relational Calculus

Normalization:-

It is a technique used in database to eliminate redundancy and solve problems in insertion, deletion and updation. It is not compulsory. There are different forms of Normalization and each form/level has some requirements. It is performed after designing logical database.

Types of Anomalies:-

- 1) Redundancy
- 2) Insertion Anomaly
- 3) Deletion Anomaly
- 4) Updation Anomaly

"Best of Luck"